

Software Requirements Specification (SRS)

Secure Messaging Web Application using USID and End-to-End Encryption (Phase 1)

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Secure Messaging Web Application developed as Phase 1. The document follows the ISO/IEC/IEEE 29148 standard.

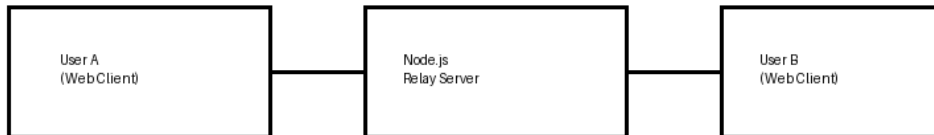
1.2 Scope

The system is a web-based secure messaging application using USID as the primary identity. Messages are end-to-end encrypted and stored only on user devices. A Node.js server acts solely as a relay.

1.3 Product Perspective

The system follows a client-server architecture with web clients and a lightweight Node.js backend.

2. System Architecture



3. System Requirement

This section describes the minimum system requirements needed to run and test the Secure Messaging Web Application in Phase 1.

Hardware Requirements

- Processor: Intel Core i3 or equivalent (or higher)
 - RAM: Minimum 4 GB
 - Storage: At least 1 GB of free disk space
 - Device Type: Desktop or Laptop computer
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Software Requirements

- Operating System: Windows, Linux, or macOS
 - Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge
 - Backend Runtime: Node.js (version 18 or higher)
 - Code Editor (Optional): Visual Studio Code or any modern IDE
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Network Requirements

- Stable internet connection
- Support for HTTPS communication
- WebSocket support for real-time message exchange

3. Data Flow Diagram (DFD)

DFD Level 0



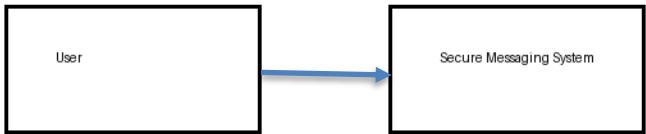
4. Sequence Diagram

Sequence Diagram



5. Use Case Diagram

Use Case Diagram



6. Functional Requirements

Req ID	Requirement	Description
FR-01	USID Generation	Generate unique cryptographic USID
FR-02	User Authentication	Authenticate users via public-key cryptography
FR-03	Encrypted Messaging	Encrypt messages using Signal Protocol
FR-04	Message Relay	Relay encrypted messages without storage
FR-05	Local Storage	Store messages only on user

		devices
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7. Non-Functional Requirements

Category	Requirement	Description
Security	E2EE	All messages encrypted end-to-end
Performance	Low Latency	Minimal message delay
Reliability	No Persistence	No server-side message storage
Usability	Simple UI	Easy to use interface

8. Requirement Traceability Matrix (RTM)

Requirement ID	Use Case	Design Component	Test Case
FR-01	User Registration	USID Module	TC-01
FR-02	Login	Auth Module	TC-02
FR-03	Send Message	Signal Encryption	TC-03
FR-04	Message Transfer	Node.js Relay	TC-04
FR-05	View Message	Local Storage	TC-05

8. Future Enhancements

- Auto-delete and burn-after-read messages
- Metadata minimization
- Forensic-resistant data wiping

9. References

1. K. Cohn-Gordon et al., "A Formal Security Analysis of the Signal Messaging Protocol," *IEEE EuroS&P*, 2017.
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7. E. Alshaer et al., "Secure Messaging for Military Applications," *Military Communications Conference*, 2022.